7 **(a)** Explain how the photoelectric effect provides evidence that supports the particulate nature of electromagnetic radiation.

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(b) The SARS-CoV-2 virus measures 140 nm in diameter.

Explain why it can be observed under an electron microscope but not an optical microscope that uses visible light.

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(c) A solar sail uses light to propel a spacecraft through space.

Explain how massless photons exert a force on a solar sail.

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